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Baffle plate for controlling wafer uniformity and methods for making the same

Abstract

Devices and methods for controlling wafer uniformity using a gas baffle plate are disclosed. In one example, a device for plasma-based processes is disclosed. The device includes: a housing defining a process chamber and a baffle plate arranged above a wafer in the process chamber. The baffle plate is configured to control plasma distribution on the wafer. The baffle plate has a shape of an annulus that comprises a first annulus sector and a second annulus sector. The first annulus sector has a first inner radius. The second annulus sector has a second inner radius that is different from the first inner radius.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a division of U.S. patent application Ser. No. 16/422,071, filed May 24, 2019, which claims priority to U.S. Provisional Patent Application No. 62/712,673 filed on Jul. 31, 2018, each of which are incorporated by reference herein in their entireties.

BACKGROUND

(1) Plasma based processing techniques have gained widespread use in fabrication of devices for various applications, such as semiconductor integrated devices, microelectronic devices, and microelectromechanical devices. When patterning techniques such as photolithography, deposition,

and etching are used to form various features on a wafer in a semiconductor process chamber, an important goal is to have uniform critical dimensions (CD) of the patterned features within the wafer.

(2) A key factor for wafer uniformity during a plasma based process, e.g. etching, deposition, or polishing, is the plasma distribution on the wafer surface. A wafer process chamber may include a gas baffle plate, a gas inlet, a gas outlet, and a radio frequency (RF) inlet. Each of these components can impact the plasma distribution in the wafer process chamber, thus impacting the CD uniformity of the wafer as well. An existing design of gas baffle plate is isotropic, i.e. treating all orientations on the plate surface to be the same, and has an inner diameter independent of the layouts of the gas inlet, the gas outlet, and the RF inlet, which cannot satisfy a uniformity requirement, especially in through-silicon via (TSV) and deep silicon etching processes which have a high standard of etching uniformity.

(3) Therefore, existing devices and methods for controlling wafer uniformity in plasma-based processes are not entirely satisfactory.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that various features are not necessarily drawn to scale. In fact, the dimensions and geometries of the various features may be arbitrarily increased or reduced for clarity of discussion. Like reference numerals denote like features throughout specification and drawings.

(2) FIG. 1 illustrates exemplary plasma flow field distributions in a cross-sectional view of a process chamber, in accordance with some embodiments of the present disclosure.

(3) FIG. 2 illustrates a cross-sectional view of an exemplary plasma-based process tool with a gas baffle plate, in accordance with some embodiments of the present disclosure.

(4) FIG. 3 illustrates an exemplary spiral inductor used in a plasma-based process tool, in accordance with some embodiments of the present disclosure.

(5) FIG. 4 illustrates a top view of an exemplary gas distribution plate (GDP) with a marked gas inlet position, in accordance with some embodiments of the present disclosure.

(6) FIG. 5 illustrates an exemplary critical dimension (CD) map of a wafer, in accordance with some embodiments of the present disclosure.

(7) FIG. 6 illustrates a top view of an exemplary baffle plate with marked positions for a gas inlet, a gas outlet, and a radio frequency (RF) inlet, in accordance with some embodiments of the present disclosure.

(8) FIG. 7 illustrates a top view of an exemplary baffle plate including multiple zones, in accordance with some embodiments of the present disclosure.

(9) FIG. 8 illustrates an exemplary method of making a baffle plate including multiple zones, in accordance with some embodiments of the present disclosure.

(10) FIG. 9 illustrates exemplary CD maps of a wafer before and after using a disclosed baffle plate, in accordance with some embodiments of the present disclosure.

(11) FIG. 10 is a flow chart illustrating an exemplary method for designing a gas baffle plate to control wafer uniformity in plasma-based process, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(12) The following disclosure describes various exemplary embodiments for implementing different features of the subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are

not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Terms such as “attached,” “affixed,” “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

(14) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(15) Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

(16) Wafer uniformity control is a tough task for all stages in plasma based processing. For example, critical dimension (CD) performance in deep silicon etching process is hard to control but is critical to final wafer testing, e.g. wafer acceptance test (WAT) or circuit probe (Cp), which can easily suffer a loop edge or a low yield. A gas baffle plate may be arranged above a wafer in a process chamber and configured to control plasma distribution on the wafer. The gas baffle plate has an inner diameter whose size can impact plasma flow field distribution on the wafer. Varying the size of the gas baffle plate can change the plasma flow field distribution, which may improve plasma uniformity and hence etching uniformity.

(17) However, due to the same inner radius along all directions of an existing gas baffle, the gas baffle does not compensate for non-uniformities in the plasma and etching caused by a layout of the process chamber. The layout may be defined by, for example, an arrangement of gas inlets, gas outlets, radio frequency (RF) inlet electrode, or a combination of the foregoing. As such, existing design of gas baffle is not enough for controlling etching uniformity. For example, etching uniformity is highly desired for notching window of a high-density deep-depth and low-pitch via, e.g. used to gain scan-through efficiency of optical sensing in an under-display fingerprint recognition component of a portable device. In one example, while silicon etching uniformity needs to be less than 10% for a through-silicon via (TSV) with a depth/width aspect ratio larger than 15 and a pitch less than 8 micrometers, a process tool with an existing design of gas baffle can merely achieve an etching uniformity of 23.3%.

(18) The present application is directed towards process devices and methods for achieving a high uniformity in plasma-based etching with a newly designed gas baffle. In some embodiments, the gas baffle has a shape of an annulus that comprises a first annulus sector having a first inner radius and a second annulus sector having a second inner radius. The first inner radius is different from the second inner radius. The values of first inner radius and the second inner radius may be

designed based on at least one of positions of the gas inlet, the gas outlet and the RF inlet. That is, the inner radius (or diameter) of the gas baffle plate is not the same along all directions, but are designed differently in different zones of the gas baffle plate based on a layout of the gas inlet, the gas outlet, and/or the RF inlet of the process chamber where the gas baffle plate is arranged.

(19) In one embodiment, the first annulus sector comprises a first point that is farthest away from the RF inlet among all points on the baffle plate; and the second annulus sector comprises a second point that is centrosymmetric to the first point around a center point of the baffle plate. In this case, the first inner radius is designed to be larger than the second inner radius to move the gas flow away from the RF inlet side of the wafer to the opposite side of the RF inlet of the wafer. This compensates for the wafer non-uniformity induced by the layout and position of the RF inlet in the process chamber.

(20) In another embodiment, the first annulus sector comprises a first point that is farthest away from the gas outlet among all points on the baffle plate; and the second annulus sector comprises a second point that is centrosymmetric to the first point around a center point of the baffle plate. In this case, the first inner radius is designed to be larger than the second inner radius to move the gas flow away from the gas outlet side of the wafer to the opposite side of the gas outlet of the wafer. This compensates for the wafer non-uniformity induced by the layout and position of the gas outlet in the process chamber.

(21) The present disclosure is applicable to wafer uniformity control during any wafer processing using a gas baffle. The disclosed gas baffle can improve etching uniformity to meet uniformity standards during bulk manufacture of TSV and deep silicon etching processes. In the present disclosure, the terms “gas baffle”, “gas baffle plate”, and “baffle plate” may be used interchangeably.

(22) FIG. 1 illustrates exemplary plasma flow field distributions **110**, **120** in a cross-sectional view of a process chamber, in accordance with some embodiments of the present disclosure. The plasma flow field distribution **110** is for a process chamber with a gas baffle plate **115** having an inner diameter of 170 mm. As in the cross-sectional view of the process chamber shown in FIG. 1, the gas baffle plate **115** has an annular shape with a center hole. The process chamber has a cathode support **116** for a wafer (not shown) to be processed. The wafer will be placed on the cathode support **116** and under the gas baffle plate **115**. During processing, the plasma gas flows from the inlets **111**, **112**, which may be holes of a GDP, into the process chamber, goes through the center hole of the gas baffle plate **115** onto the wafer, and is pumped out of the process chamber through the gas outlet **119**. A gas baffle may be employed with any plasma-based process in which uniform plasma is desired, e.g. plasma-based etching, plasma activation, etc. For simplicity, plasma-based etching will be used as an example in the following description about wafer uniformity improvement.

(23) The plasma flow field distribution **120** is for a process chamber with a gas baffle plate **125** having an inner diameter of 215 mm. As in the cross-sectional view of the process chamber shown in FIG. 1, the gas baffle plate **125** has an annular shape with a center hole. The process chamber has a cathode support **126** for a wafer (not shown) to be processed. The wafer will be placed on the cathode support **126** and under the gas baffle plate **125**. During processing, the plasma gas flows from the inlets **121**, **122**, which may be holes of a GDP, into the process chamber, goes through the center hole of the gas baffle plate **125** onto the wafer, and is pumped out of the process chamber through the gas outlet **129**.

(24) As shown in FIG. 1, the gas baffle plate **115** having a smaller inner diameter of 170 mm makes the plasma flow field more focused at the inner side of the wafer; while the gas baffle plate **125** having a larger inner diameter of 215 mm enables the plasma flow field more extended to the outer side of the wafer. A larger inner diameter of a gas baffle may improve wafer etching rate uniformity, but a too large inner diameter of a gas baffle may make the average etching rate of the wafer to be low. While the inner diameter size of an existing gas baffle is merely based on a

customer design requirement for the semiconductor wafer, the disclosed gas baffle has an inner diameter size designed based on positions of the gas inlet, the gas outlet, and/or the RF inlet of the process chamber as well.

(25) FIG. 2 illustrates a cross-sectional view of an exemplary plasma-based process tool **200** with a gas baffle plate **270**, in accordance with some embodiments of the present disclosure. The process tool **200** may be configured to perform plasma-based etching, such as, for example, deep reactive ion etching (DRIE) or plasma etching. In some embodiments, the process tool **200** is configured to perform a Bosch process. As shown in FIG. 2, the processing tool **200** includes a housing **210** defining a process chamber, and a GDP **240** arranged in the process chamber **210**. The housing **210** comprises a gas inlet **220** configured to receive a process gas **252**, and a gas outlet **230** configured to expel the processed gas. The GDP **240** is configured to distribute the process gas **252** within the process chamber **210**.

(26) In one embodiment, the GDP **240** has a plurality of holes evenly distributed thereon. The term “even” or “evenly” herein may refer to a uniform distribution of the holes with a constant density in a given area. After the GDP **240** receives the process gas **252** from the gas inlet **220**, the process gas **252** enters the process chamber **210** through the holes on the GDP **240**. The process gas **252** may include, for example, sulfur hexafluoride (SF₆) and/or octofluorocyclobutane (C₄F₈). As such, the GDP **240** distributes the process gas **252** received from the gas inlet **220** into the process chamber **210** through holes of the GDP **240**.

(27) The GDP **240** is located on top of an upper region of the process chamber **210** that is on top of a lower region of the process chamber **210** along the Z direction. The lower region of the process chamber **210** accommodates a wafer support **280** and is connected to a pumping line **232** through a gas outlet **230** of the housing. The wafer support **280** is configured to support a wafer **290** and, in some embodiments, is or otherwise comprises an electrode. The wafer **290** may be, for example, a 350 millimeter or 450 millimeter semiconductor wafer. The electrode may be, for example, electrically coupled to an RF source configured to promote the migration of particles from overlying plasma **254** towards the wafer support **280**. In one embodiment, the pumping line **232** is connected to an exhaust pump (not shown in FIG. 2) configured to remove gases **250**, **254** from the process chamber **310** and/or to otherwise control a pressure of the process chamber **310** relative to an ambient environment of the process tool **200**.

(28) As shown in FIG. 2, the process chamber **210** further comprises a spiral inductor **260** laterally spirals around the upper region of the process chamber and is electrically coupled to an RF source **262**. The spiral inductor **260** is configured to receive RF energy from the RF source **262** through an RF inlet (not shown in FIG. 2) and excite the process gases **252** using the RF energy, thereby producing the plasma **254** with a high density in the upper region of the process chamber **210**.

(29) A gas baffle **270** having an annular shape is placed above the wafer **290** to adjust the plasma distribution on the wafer **290**. In operation, the plasma **254** goes through the hole of the gas baffle **270** and interfaces with the wafer **290** on the wafer support **280** to perform plasma-based etching. For example, the plasma **254** may chemically react with the wafer **290** to remove material from the wafer **290**. As another example, chemical reaction of the wafer **290** with the plasma **254** and bombardment of the wafer **290** with particles of the plasma **254** may be employed to remove material from the wafer **290**.

(30) The plasma distribution and hence etching rate distribution on the wafer **290** can be controlled by the gas baffle plate **270** through a design of the structure and inner diameters (or inner radii) of the gas baffle plate **270**, based on position(s) of the gas inlet **220**, the gas outlet **230**, and/or the RF inlet of the spiral inductor **260**. In this example, the gas outlet **230** is located on the left side, i.e. to the X direction, of the gas baffle plate **270**, while the gas inlet **220** is located towards the right side, i.e. towards the -X direction, of the gas baffle plate **270**. According to various embodiments, the gas inlet **220** and/or the gas outlet **230** may be located at other locations relative to the gas baffle plate **270**. The RF inlet of the spiral inductor **260** is not shown in the cross-sectional view of the

process tool **200** in FIG. 2. Since the gas outlet **230** and the pumping line **232** are coupled to each other, a position of the gas outlet **230** corresponds to a position of the pumping line **232**. In the following description, a position of a gas outlet will be used to refer to both positions of the gas outlet and the connected pumping line.

(31) FIG. 3 illustrates a top view **300** of an exemplary spiral inductor **310** used in a plasma-based process tool, in accordance with some embodiments of the present disclosure. As shown in FIG. 3, the spiral inductor **310** has a powered end **320** and a grounded end **330**. The powered end **320** is coupled to an RF source and serves as an RF inlet to receive RF energy from the RF source. Compared to other portions of the spiral inductor **310**, the RF inlet **320** is a portion that has a higher voltage and a higher ionization energy, which induces a stronger magnetic field and higher plasma density at the RF inlet **320**. As such, a design of a gas baffle plate above the spiral inductor **310** may take into consideration of the position of the RF inlet **320**. In this embodiment, the RF inlet **320** is located at the upper left side, i.e. to a direction between X and -Y, of the spiral inductor **310**. The RF inlet **320** may be located at another direction of the spiral inductor **310** according to other embodiments.

(32) FIG. 4 illustrates a top view of an exemplary GDP **400** with a marked gas inlet position, in accordance with some embodiments of the present disclosure. In one embodiment, the GDP **400** may be implemented as the GDP **240** in FIG. 2. As shown in FIG. 4, the GDP **400** in this example has an annular shape bordered by an outer circle **202** and an inner circle **204**. The GDP **400** has a plurality of holes **405** evenly distributed thereon. As discussed before, a gas inlet is located above, i.e. to the Z direction of, the GDP **400** to receive process gas. FIG. 4 shows a projection area **410** of the gas inlet onto the GDP **400**. The projection area **410** is a GDP portion having a shortest distance to the gas inlet and serves as a process gas source for the GDP **400**, and is referred to as the gas inlet area **410**. That is, the process gas received by the gas inlet first arrives at the gas inlet area **410** of the GDP **400**, and then goes into the process chamber through the holes **405**.

(33) As shown in FIG. 4, the process gas may move along two paths **432**, **434** into the process chamber. The path **432** extends from the gas inlet area **410** to the left side (along the -Y direction) and then to the upper side (along the X direction) of the gas inlet area **410**. The path **434** extends from the gas inlet area **410** to the right side (along the Y direction) and then to the upper side (along the X direction) of the gas inlet area **410**. In the example of FIG. 4, a first wafer portion under the gas inlet area **410** would interface with a higher density of process gas, e.g. C.sub.4F.sub.8, than the gas density at a second wafer portion under the area **420** which is farthest away from the gas inlet area **410** on the GDP **400**, which would decrease the etching rate at the first wafer portion and increase the etching rate at the second wafer portion. As such, position of the gas inlet area **410** can impact the wafer uniformity as well.

(34) FIG. 5 illustrates an exemplary critical dimension (CD) map **500** of a wafer **510**, in accordance with some embodiments of the present disclosure. As shown in FIG. 5, the CD map **500** shows a distribution of CD performance, e.g. bulk chemical distribution (BCD), on the wafer **510**. Based on a top view of the wafer **510**, FIG. 5 shows a corresponding projection of the gas baffle plate placed above the wafer **510**, where the gas baffle plate has an annular shape bordered by an outer circle **522** and an inner circle **524**. In addition, FIG. 5 also shows wafer portions of the wafer **510** corresponding to the gas inlet, the gas outlet, and the RF inlet. The grey scale shown in FIG. 5 reflects critical dimensions.

(35) In this example, a wafer portion **540** corresponds to the gas inlet. That is, the wafer portion **540** is closer to the gas inlet than any other portion of the wafer **510**, and is referred to as the gas inlet portion **540**. In this example, the gas inlet portion **540** is located at the right lower corner, to a direction between Y and -X, of the wafer **510**. As discussed before, due to a higher process gas density at the gas inlet portion **540**, the gas inlet portion **540** tends to have a lower etching rate than other wafer portions. As such, with a gas baffle plate having a same diameter along all directions of the plate surface, the BCD performance of a point on the wafer **510** tends to become lower as the

point moves closer to the gas inlet portion **540**.

(36) In this example, a wafer portion **550** corresponds to the RF inlet. That is, the wafer portion **550** is closer to the RF inlet than any other portion of the wafer **510**, and is referred to as the RF inlet portion **550**. In this example, the RF inlet portion **550** is located at the left upper portion, to a direction between X and -Y, of the wafer **510**. As discussed before, due to a stronger magnetic field and a higher plasma density at the wafer portion **550**, the RF inlet portion **550** tends to have a higher etching rate than other wafer portions. So with a gas baffle plate having a same diameter along all directions of the plate surface, the BCD performance of a point on the wafer **510** tends to become higher as the point moves closer to the RF inlet portion **550**.

(37) In this example, a wafer portion **530** corresponds to the gas outlet. That is, the wafer portion **530** is closer to the gas outlet than any other portion of the wafer **510**, and is referred to as the gas outlet portion **530**. In this example, the gas outlet portion **530** is located at the top portion, to the X direction, of the wafer **510**. As such, process gas received from the gas inlet can reach the gas outlet portion **530** following either a shorter path along the right side of the gas inlet portion **540** or a longer path along the left side of the gas inlet portion **540**. As shown in FIG. 5, with a gas baffle plate having a same diameter along all directions of the plate surface, the etching rate tends to be lower at the right side of the wafer **510** compared to the etching rate at the left side of the wafer **510**.

(38) FIG. 6 illustrates a top view of an exemplary baffle plate **600** with marked positions for a gas inlet, a gas outlet, and a radio frequency (RF) inlet, in accordance with some embodiments of the present disclosure. In one embodiment, the baffle plate **600** may be implemented as the gas baffle plate **270** in FIG. 2. As shown in FIG. 6, the gas baffle plate **600** in this example has an annular shape bordered by an outer circle **622** and an inner circle **624**. Based on the top view of the gas baffle plate **600**, FIG. 6 shows a corresponding projection of the wafer **610** placed under the gas baffle plate **600**.

(39) In one embodiment, the wafer **610** and the gas baffle plate **600** are the same as or correspond to the wafer **510** and the gas baffle plate on the wafer **510**, respectively. In addition, FIG. 6 also shows gas baffle portions of the gas baffle plate **600** corresponding to the gas inlet, the gas outlet, and the RF inlet.

(40) In this example, a gas baffle portion **640** corresponds to the gas inlet. That is, the gas baffle portion **640** is closer to the gas inlet than any other portion of the gas baffle plate **600**, and is referred to as the gas inlet portion **640**. In this example, the gas inlet portion **640** is located at the right lower corner, to a direction between Y and -X, of the gas baffle plate **600**. As discussed before, due to a higher process gas density under the gas inlet portion **640**, a wafer portion under the gas inlet portion **640** tends to have a lower etching rate than other wafer portions.

(41) In this example, a gas baffle portion **650** corresponds to the RF inlet. That is, the gas baffle portion **650** is closer to the RF inlet than any other portion of the gas baffle plate **600**, and is referred to as the RF inlet portion **650**. In this example, the RF inlet portion **650** is located at the left upper portion, to a direction between X and -Y, of the gas baffle plate **600**. As discussed before, due to a stronger magnetic field and a higher plasma density under the gas baffle plate **600**, a wafer portion under the RF inlet portion **650** tends to have a higher etching rate than other wafer portions.

(42) In this example, a gas baffle portion **630** corresponds to the gas outlet. That is, the gas baffle portion **630** is closer to the gas outlet than any other portion of the gas baffle plate **600**, and is referred to as the gas outlet portion **630**. In this example, the gas outlet portion **630** is located at the top portion, to the X direction, of the gas baffle plate **600**. As such, process gas received from the gas inlet can reach the gas outlet portion **630** following either a shorter path **664** along the right side of the gas inlet portion **640** or a longer path **662** along the left side of the gas inlet portion **640**.

(43) As shown in FIG. 6, the baffle plate **600** includes a point **661** that is farthest away from the gas outlet portion **630** among all points on the baffle plate **600**; and a point **663** that is farthest away

from the RF inlet portion **650** among all points on the baffle plate **600**. The baffle plate **600** also includes a point **662** that is centrosymmetric to the point **661** around a center point **601** of the baffle plate **600**; and a point **664** that is centrosymmetric to the point **663** around the center point **601** of the baffle plate **600**. According to the CD performance in FIG. 5, CD performance tends to be lower between the point **661** and the point **663** than other portions. As such, a special structure (e.g. a larger inner radius) may be designed for a first portion between the point **661** and the point **663**, to make the plasma flow move to the first portion from a second portion between the point **662** and the point **664**. In one embodiment, the special structure may be designed for an extended portion between the point **665** and the point **666**. The extended portion may be achieved by extending 0~60 degrees from the first portion. In one embodiment, the point **665** is to the left (along the -Y direction) from the point **661** by a degree that is in a range from zero to about 30 degrees; and the point **666** is to the right (along the Y direction) from the point **663** by a degree that is in a range from zero to about 30 degrees.

(44) FIG. 7 illustrates a top view of an exemplary baffle plate **700** including multiple zones, in accordance with some embodiments of the present disclosure. In one embodiment, the baffle plate **700** may be implemented as the baffle plate **270** in FIG. 2. As shown in FIG. 7, the baffle plate **700** in this example has an annular shape bordered by an outer circle **722** and an inner circle **724**. Based on the top view of the gas baffle plate **700**, FIG. 7 shows a corresponding projection of the wafer **710** placed under the gas baffle plate **700**. Similar to FIG. 6, FIG. 7 shows baffle plate portions of the baffle plate **700** corresponding to the gas inlet, the gas outlet, and the RF inlet, including: a baffle plate portion **740** corresponding to the gas inlet that is closer to the gas inlet than any other portion of the baffle plate **700**, and is located at the right lower corner, to a direction between Y and -X, of the baffle plate **700**, and is referred to as the gas inlet portion **740**; a baffle plate portion **750** corresponding to the RF inlet that is closer to the RF inlet than any other portion of the baffle plate **700** and is located at the left upper portion, to a direction between X and -Y, of the baffle plate **700**, and is referred to as the RF inlet portion **750**; and a baffle plate portion **730** corresponding to the gas outlet that is closer to the gas outlet than any other portion of the baffle plate **700** and is located at the top portion, to the X direction, of the baffle plate **700**, and is referred to as the gas outlet portion **730**.

(45) As shown in FIG. 7, the baffle plate **700** includes a point **761** that is farthest away from the gas outlet portion **730** among all points on the baffle plate **700**; and a point **763** that is farthest away from the RF inlet portion **750** among all points on the baffle plate **700**. The baffle plate **700** also includes a point **762** that is centrosymmetric to the point **761** around a center point **701** of the baffle plate **700**; and a point **764** that is centrosymmetric to the point **763** around the center point **701** of the baffle plate **700**. In this embodiment, the baffle plate **700** is divided into a plurality of zones: a first zone **770** including a first annulus sector, a second zone **780** including a second annulus sector, and a third zone **790** including a third annulus sector. The first annulus sector **770** comprises a point **763** that is farthest away from the RF inlet among all points on the baffle plate **700**, and a point **761** that is farthest away from the gas outlet among all points on the baffle plate **700**. The second annulus sector **780** comprises a point **764** that is centrosymmetric to the point **763** around the center point **701** of the baffle plate **700**, and a point **762** that is centrosymmetric to the point **761** around the center point **701** of the baffle plate **700**.

(46) In the example shown in FIG. 7, the first annulus sector **770** comprises a fourth annulus sector **772** that is bordered by two straight edges and two arcs. A first edge of the two straight edges comprises the point **763** that is farthest away from the RF inlet among all points on the baffle plate **700**. A second edge of the two straight edges comprises the point **761** that is farthest away from the gas outlet among all points on the baffle plate **700**. The first annulus sector **770** is also bordered by two straight edges and two arcs, where a first edge of the two straight edges comprises the point **765** to the left (along the -Y direction) of the point **761**, and a second edge of the two straight edges comprises the point **766** to the right (along the Y direction) of the point **763**. In one

embodiment, a central angle of the first annulus sector **770** is larger than a central angle of the fourth annulus sector **772** by a degree that is in a range from zero to about 60 degrees. In one embodiment, the point **765** is to the left (along the $-Y$ direction) from the point **761** by a degree that is in a range from zero to about 30 degrees; and the point **766** is to the right (along the Y direction) from the point **763** by a degree that is in a range from zero to about 30 degrees. In one embodiment, the fourth annulus sector **772** is located at a center of the first annulus sector **770**. In another embodiment, the fourth annulus sector **772** comprises the gas inlet portion **740** or a point that is closest to the gas inlet among all points on the baffle plate **700**.

(47) According to the CD performance in FIG. 5, CD performance tends to be lower in the first annulus sector **770**, and especially in the fourth annulus sector **772**. As such, a special structure (e.g. a longer inner radius or diameter) may be designed for the first annulus sector **770**, to make the plasma flow move to from the second annulus sector **780** to the first annulus sector **770**. The baffle plate **700** has a diameter D1 **705** connecting the first annulus sector **770** and the second annulus sector **780**; and a diameter D2 **706** cross the third annulus sector **790** and the center point **701**. In one embodiment, the diameter D1 **705** is longer than the diameter D2 **706**, e.g. by about a percentage that is in a range from about 1% to about 16%. In one embodiment, the diameter D1 **705** is along a first direction from the center point **701** to a point between the point **761** and the point **763**, or from the center point **701** to a point that has a minimum average distance from the point **761** and the point **763** among all points on the baffle plate **700**. The diameter D1 **705** along the first direction is longer than a diameter along a second direction that is orthogonal to the first direction. This can compensate for the wafer non-uniformity induced by at least one of positions of the gas inlet, the gas outlet and the RF inlet relative to the baffle plate **700**.

(48) FIG. 8 illustrates an exemplary method of making a baffle plate **820** including multiple zones, in accordance with some embodiments of the present disclosure. FIG. 8 shows a baffle plate **810** having a uniform inner radius R **812**, and a baffle plate **820** having two inner radii R1 **825** and R2 **826**. The baffle plate **810** in this example has an annular shape bordered by an outer circle **813** and an inner circle **814**. The inner radius R **812** connecting a center point **811** of the baffle plate **810** to the inner circle **814** is the same along all directions on the plate surface of the baffle plate **810**.

(49) The baffle plate **820** in this example has an annular shape bordered by an outer circle **822** and two inner arcs **823**, **824**. The inner radius R1 **825** connects a center point **821** of the baffle plate **820** to the inner arc **823**; and the inner radius R2 **826** connects the center point **821** of the baffle plate **820** to the inner arc **824**. As shown in FIG. 8, the baffle plate **820** includes a first annulus sector **870** having the inner radius R2 **826** and a second annulus sector **890** having the inner radius R1 **825**. The first annulus sector **870** and the second annulus sector **890** share the same **822** and have a same outer radius. The first annulus sector **870** is bordered by the inner arc **824**, the outer circle **822**, and two straight edges that comprise points **865**, **866** respectively. In one embodiment, the point **865** is at a 7 o'clock direction from the center point **821**; and the point **866** is at a 3 o'clock direction from the center point **821**. The second annulus sector **890** is to the left (along the $-Y$ direction) of the first annulus sector **870**. In some embodiments, the inner curvature of the first annulus sector **870** is the same as the outer curvature of the first annulus sector **870**. In other embodiments, the inner curvature of the first annulus sector **870** is different from the outer curvature of the first annulus sector **870**.

(50) In one embodiment, the baffle plate **820** may be implemented as the baffle plate **270** in FIG. 2. In one embodiment, the baffle plate **810** and the baffle plate **820** may be arranged in a process chamber with the positions of the gas inlet, the gas outlet, and the RF inlet same as those marked in FIGS. 5-7. Accordingly, the first annulus sector **870** includes a first point that is farthest away from the gas outlet among all points on the baffle plate **820**; and a second point that is farthest away from the RF inlet among all points on the baffle plate **820**.

(51) In one embodiment, the inner radius R1 **825** is designed to be larger than the uniform inner radius R **812** of the baffle plate **810** to improve wafer uniformity by extending the plasma flow

towards outer side of the wafer. For example, while the uniform inner radius R **812** is equal to about 85 mm based on a wafer type to be processed, the inner radius R1 **825** is increased to about 95 mm for the same wafer type.

(52) In addition, the inner radius R2 **826** is longer than the inner radius R1 **825** by a length L **828**, to compensate for the wafer non-uniformity induced by at least one of positions of the gas inlet, the gas outlet and the RF inlet relative to the baffle plate **820**. The length L **828** may be in a range from about 1 mm to about 15 mm, based on an inner radius (R1 and R2) of about 100 mm, to compensate for the wafer non-uniformity. In one embodiment, the inner radius R2 **826** is longer than the inner radius R1 **825** by a percentage that is in a range from about 1% to about 16%, which is not too small to differentiate R1 and R2 and not too large to over-compensate. For example, while the inner radius R1 **825** is equal to about 95 mm, the inner radius R2 **826** is equal to about 103 mm, longer than the inner radius R1 **825** by about 8 mm. Having a larger inner radius, the first annulus sector **870** attracts more plasma flow from the opposite side of the baffle plate **820**, which further improves the wafer uniformity.

(53) FIG. **9** illustrates exemplary CD maps of a wafer before and after using a disclosed baffle plate, e.g. the baffle plate disclosed in any of FIGS. **1-8**, in accordance with some embodiments of the present disclosure. As shown in FIG. **9**, the CD performance of a wafer processed using a disclosed baffle plate can achieve a higher uniformity than that of a wafer processed without using the disclosed baffle plate. In one example, a TSV BCD distribution on the wafer before using the disclosed baffle plate has a mean of about 1034, a 3-sigma value of about 465, and a uniformity of 23.3%; while a TSV BCD distribution on the wafer after using the disclosed baffle plate has a same mean of about 1034, a smaller 3-sigma value of about 206, and an improved uniformity of 12.6%. The grey scale shown in FIG. **9** reflects critical dimensions.

(54) FIG. **10** is a flow chart illustrating an exemplary method **1000** for designing a gas baffle plate to control wafer uniformity in plasma-based processes, in accordance with some embodiments of the present disclosure. At operation **1002**, a first position of a gas outlet of a process chamber is determined. The gas outlet is configured to expel a processed gas from the process chamber. At operation **1004**, a second position of an RF inlet configured to receive RF energy for generating plasma in the process chamber is determined. At operation **1006**, the baffle plate is refined based on at least one of the first position and the second position. The baffle plate has an annular shape that comprises a first annulus sector having a first inner radius and a second annulus sector having a second inner radius that is different from the first inner radius. The order of the operations shown in FIG. **10** may be changed according to different embodiments of the present disclosure.

(55) In an embodiment, a device for plasma-based processes is disclosed. The device includes: a housing defining a process chamber and a baffle plate arranged above a wafer in the process chamber. The baffle plate is configured to control plasma distribution on the wafer. The baffle plate has a shape of an annulus that comprises a first annulus sector and a second annulus sector. The first annulus sector has a first inner radius. The second annulus sector has a second inner radius that is different from the first inner radius.

(56) In another embodiment, a process chamber is disclosed. The process chamber includes: a gas inlet configured to receive a process gas; a gas outlet configured to expel a processed gas; a radio frequency (RF) inlet configured to receive RF energy for exciting the process gas to generate plasma in the process chamber; and a baffle plate located in the process chamber and configured to control plasma distribution in the process chamber. The baffle plate has a hole with a diameter that varies along different directions based on at least one of positions of the gas inlet, the gas outlet and the RF inlet.

(57) In yet another embodiment, a method for designing a baffle plate is disclosed. The method includes: determining a first position of a gas outlet of a process chamber, wherein the gas outlet is configured to expel a processed gas from the process chamber; determining a second position of a radio frequency (RF) inlet for the process chamber, wherein the RF inlet is configured to receive

RF energy for generating plasma in the process chamber; and refining the baffle plate based on at least one of the first position and the second position. The baffle plate is to be arranged above a wafer in the process chamber and configured to control plasma distribution on the wafer. The baffle plate has a shape of an annulus that comprises a first annulus sector having a first inner radius and a second annulus sector having a second inner radius. The first inner radius is different from the second inner radius based on the refining.

(58) The foregoing outlines features of several embodiments so that those ordinary skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for making a process chamber having a baffle plate therein, the method comprising: determining a first position of a gas outlet of a process chamber, wherein the gas outlet is configured to expel a processed gas from the process chamber; determining a second position of a radio frequency (RF) inlet for the process chamber, wherein the RF inlet is configured to receive RF energy for generating plasma in the process chamber; and refining the baffle plate based on at least one of the first position and the second position, wherein: the baffle plate is to be arranged above a wafer in the process chamber and configured to control plasma distribution on the wafer, the baffle plate has a shape of an annulus that comprises a first annulus sector having a first inner radius and a second annulus sector having a second inner radius, and the first inner radius is different from the second inner radius based on the refining.
2. The method of claim 1, wherein: the first annulus sector comprises a third annulus sector that is bordered by two straight edges and two arcs; a first edge of the two straight edges comprises a first point that is farthest away from the RF inlet among all points on the baffle plate; a second edge of the two straight edges comprises a second point that is farthest away from the gas outlet among all points on the baffle plate; and refining the baffle plate comprises increasing the first inner radius to be larger than the second inner radius.
3. The method of claim 2, wherein: refining the baffle plate comprises increasing a central angle of the first annulus sector to be larger than a central angle of the third annulus sector by a degree that is in a range from zero to about 60 degrees.
4. The method of claim 2, wherein increasing the first inner radius comprises at least one of: increasing the first inner radius to be longer than the second inner radius by a length that is in a range from about 1 mm to about 15 mm; and increasing the first inner radius to be longer than the second inner radius by a percentage that is in a range from about 1% to about 16%.
5. The method of claim 1, further comprising determining a third position of a gas inlet of the process chamber, wherein: the gas inlet is configured to receive the process gas; and the baffle plate is refined based on the third position as well.
6. A method for making a process chamber having a baffle plate therein, the method comprising: providing a process chamber; arranging a baffle plate in the process chamber so that the baffle plate is configured to be positioned above a wafer within the process chamber and control plasma distribution on the wafer; and providing a radio frequency (RF) inlet on the process chamber, the RF inlet configured to receive RF energy for generating plasma for processing the wafer, wherein: a top flat surface of the baffle plate has an annulus shape, the annulus shape of the baffle plate consists of: a first annulus sector and a second annulus sector that are both on the top flat surface,

the first annulus sector has a first inner radius, the first annulus sector comprises a first point that is farthest away from the RF inlet among all points on the top flat surface of the baffle plate, and the second annulus sector has a second inner radius that is smaller than the first inner radius.

7. The method of claim 6, further comprising: providing a gas inlet to the process chamber, the gas inlet configured to receive a process gas; and providing a gas outlet to the process chamber, the gas outlet configured to expel a processed gas.

8. The method of claim 6, wherein the second annulus sector comprises a second point that is centrosymmetric to the first point around a center point of the baffle plate.

9. The method of claim 6, wherein: the first annulus sector comprises a third point that is farthest away from the gas outlet among all points on the top flat surface of the baffle plate; and the second annulus sector comprises a fourth point that is centrosymmetric to the third point around a center point of the baffle plate.

10. The method of claim 6, wherein: the first annulus sector comprises a third annulus sector that is bordered by two straight edges and two arcs; a first edge of the two straight edges comprises the first point; and a second edge of the two straight edges comprises the third point.

11. The method of claim 10, wherein: a central angle of the first annulus sector is larger than a central angle of the third annulus sector by a degree that is in a range from zero to about 60 degrees.

12. The method of claim 10, wherein the third annulus sector is located at a center of the first annulus sector.

13. The method of claim 10, wherein the third annulus sector comprises a fifth point that is closest to the gas inlet among all points on the top flat surface of the baffle plate.

14. The method of claim 6, wherein: the second inner radius is about 95 mm; and the first inner radius is longer than the second inner radius by a length that is in a range from about 1 mm to about 15 mm.

15. The method of claim 6, wherein the first inner radius is longer than the second inner radius by a percentage that is in a range from about 1% to about 16%.

16. The method of claim 6, wherein the annulus is formed by the first annulus sector and the second annulus sector.

17. The method of claim 6, wherein the first annulus sector and the second annulus sector have a same outer radius.

18. A method of making a process chamber, comprising: providing a gas inlet configured to receive a process gas; providing a gas outlet configured to expel a processed gas; providing a radio frequency (RF) inlet configured to receive RF energy for exciting the process gas to generate plasma in the process chamber; and providing a baffle plate located in the process chamber and configured to control plasma distribution in the process chamber, wherein a top flat surface of the baffle plate has a central hole with a diameter that varies along different directions on the top flat surface based on at least one of: a first position of the gas inlet, a second position of the gas outlet, or a third position of the RF inlet, and an outer edge of the baffle plate on the top flat surface has a circular shape whose center point is within and part of the central hole.

19. The method of claim 18, wherein: the baffle plate comprises: a first point that is farthest away from the RF inlet among all points on the top flat surface of the baffle plate, a second point that is farthest away from the gas outlet among all points on the top flat surface of the baffle plate, and a third point that has a minimum average distance from the first point and the second point among all points on the baffle plate; and the diameter is longer along a first direction crossing the third point than that along a second direction that is orthogonal to the first direction.

20. The method of claim 18, wherein the diameter is longer along the first direction than that along the second direction by at least one of: a length that is in a range from about 1 mm to about 15 mm; or a percentage that is in a range from about 1% to about 16%.
